Progress Report Team 7

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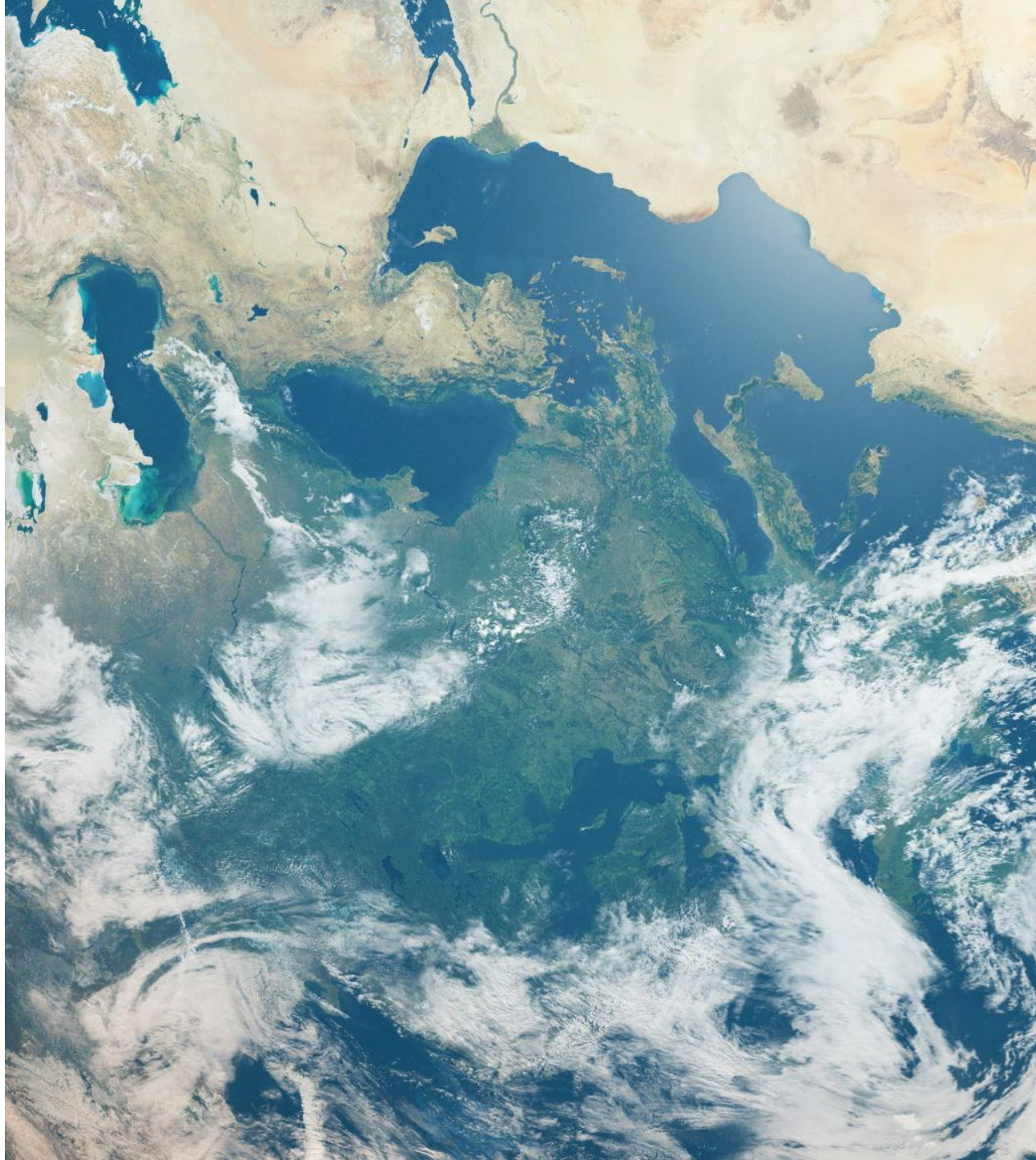
Research Goals

- **Research Topic:** Spatial distribution and concentration patterns of Rare Earth Elements (REE) on a global scale.

- **Significance:** REE are strategic minerals for modern technology; understanding their natural clustering is vital for resource management.

Hypotheses

- **Hypothesis 1 (Geospatial Clustering):** Do global REE occurrences form distinct spatial clusters that correlate with specific tectonic plate boundaries?
- **Hypothesis 2 (Compositional Diversity):** Is there a significant difference in the proportion of light rare earth and heavy rare earth minerals among different clustering groups?
- **Analysis Design:** Extracting coordinates and mineral species via API, followed by data cleaning and clustering analysis.



Task Allocation

Name	Task
Qiwei Liang	Data Engineer: Handling the Mindat API script and cleaning up the raw data.
Mengna Cheng	GIS/Visualizer: Making the maps and all the cool charts for the results.
Michael King	ML Developer: Building the Clustering model and tuning it.
Rohit Narwal	Technical Writer: Taking the lead on the 30% written report (4b).

Data Management Plan

Data Source: Mineral data will be collected from the Mindat API

Data Storage: Data will be stored in CSV/JSON format and managed using GitHub

Data Cleaning: Missing values and inconsistent entries will be handled during preprocessing

Data Processing: Relevant features (e.g., location, mineral type) will be selected for clustering analysis

Data Sharing: Cleaned datasets and results will be shared among team members via GitHub and SharePoint

Project Timeline

Immediate Goal: Complete the global data pull for REE occurrences.

Next Steps:

- Exploratory Data Analysis (EDA) to calculate cluster proportions.
- ML model training and final visualization.

Challenges

- Data from Mindat API may be incomplete or inconsistent
- Difficulty in cleaning and preprocessing geological data
- Choosing appropriate features for clustering analysis
- Limited experience with clustering algorithms and visualization tools

Thank you!